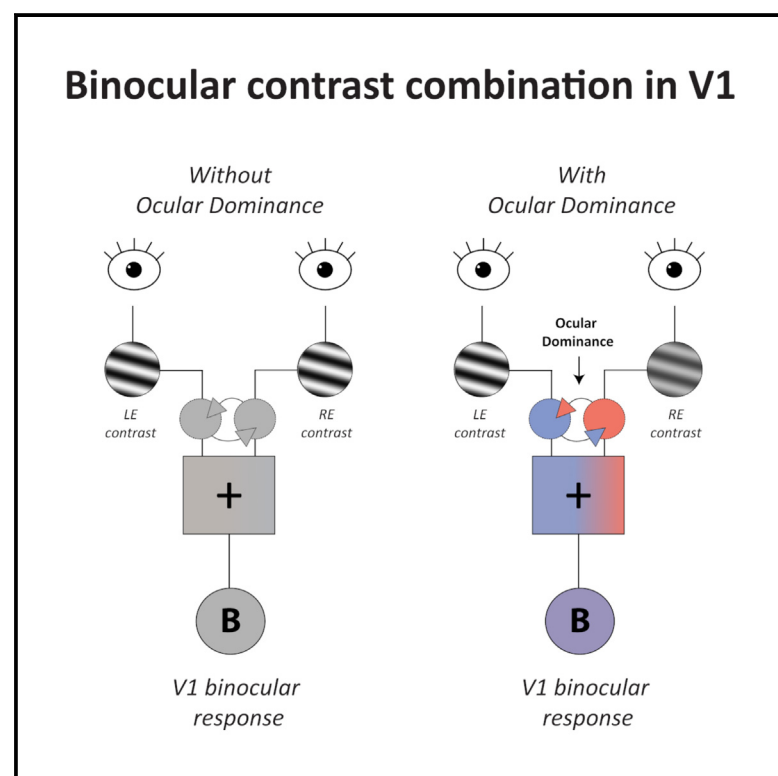


A role for ocular dominance in binocular integration

Graphical abstract



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In brief

Some cortical neurons exhibit a response preference for one eye or the other. Mitchell et al. report that this ocular dominance (OD) indicates the level of gain control exerted on the non-preferred eye during binocular contrast combination. Incorporating OD into models of binocular contrast combination improves model performance.

Highlights

- V1 dichoptic responses are weighted more by contrast placed in the dominant eye
- Asymmetry of V1 dichoptic responses correlates with neurons' ocular dominance
- Ocular dominance can be parameterized within a divisive normalization framework
- Interocular normalization weighted by OD best explains V1 dichoptic responses



Article

A role for ocular dominance in binocular integration

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SUMMARY

Neurons in the primate primary visual cortex (V1) combine left- and right-eye information to form a binocular output. Controversy surrounds whether ocular dominance, the preference of these neurons for one eye over the other, is functionally relevant. Here, we demonstrate that ocular dominance impacts gain control during binocular combination. We recorded V1 spiking activity while monkeys passively viewed grating stimuli. Gratings were either presented to one eye (monocular), both eyes with the same contrasts (binocular balanced), or both eyes with different contrasts (binocular imbalanced). We found that contrast placed in a neuron's dominant eye was weighted more strongly than contrast placed in a neuron's non-dominant eye. This asymmetry covaried with neurons' ocular dominance. We then tested whether accounting for ocular dominance within divisive normalization improves the fit to neural data. We found that ocular dominance significantly improved model performance, with interocular normalization providing the best fits. These findings suggest that V1 ocular dominance is relevant for response normalization during binocular stimulation.

INTRODUCTION

In the primate primary visual cortex (V1), most neurons respond to retinal stimulation in either eye.¹ That is, most V1 neurons are considered *binocular*. However, V1 neurons tend to express a response preference for one eye or the other, or *ocular dominance* (OD). This phenomenon is tied to the segregation of eye-specific thalamocortical afferents to the recipient layers of V1, forming *OD columns*.^{2–4} OD columns exist in a wide range of species, typically in those that have front-facing eyes and overlapping visual fields.⁵ They have been demonstrated physiologically^{1,3,6} as well as anatomically.^{7–11} Yet, while the existence of V1 OD is indisputable, its significance with respect to visual processing remains a matter of debate.¹²

Interest in the function of OD surged when it was discovered that V1 neurons are sensitive to (tuned for) retinal disparity.^{13–15} Subsequent experiments reported a relationship between disparity tuning of V1 neurons and OD strength,¹⁶ suggesting that V1 neurons utilize an imbalance of ocular inputs to compute depth.¹⁷ However, this finding has been challenged and, in some cases, overturned by improvements in electrophysiological technique and reduced measurement error.^{12,18} Based on these experiments, it is not unreasonable to view the physiological property of OD as extraneous with respect to binocular processing.

In recent years, evidence has been accumulating in favor of an alternative view. Recent studies have demonstrated a relationship between the physiological property of OD and V1 binocular response formation.^{19,20} For example, neurons dominated strongly by one eye in the input layers of the V1 (i.e., seemingly monocular neurons) have been shown to modulate under binocular stimulation in a manner that is predicted by the strength of

their OD.^{19,20} OD has also been implicated in computational models of V1 binocular cells. For example, a recent re-analysis of data from anesthetized cats suggested a role for OD in controlling monocular gain before binocular summation.²¹ Taken together, these findings suggest that a neuron's monocular selectivity for eye might be more relevant for early-stage binocular processing than previously thought.

Here, we investigate whether OD is relevant for binocular contrast combination in the primate V1. To do so, we recorded from V1 OD columns using linear microelectrode arrays while macaque monkeys viewed grating stimuli, either monocularly or binocularly, through a mirror stereoscope. We collected spiking responses to these stimuli from V1 units across a wide array of OD, ranging from purely monocular to fully binocular neurons. We found that placing higher contrast in a neuron's dominant eye was more influential for binocular response formation than the same stimulus placed in the non-dominant eye. This asymmetry scaled with OD. Using this knowledge, we found that accounting for OD in computational models of binocular contrast combination significantly improved performance. Together, our results demonstrate that OD is relevant for binocular contrast combination by contributing to interocular gain-control.

RESULTS

We were interested to determine whether OD is relevant for binocular contrast combination in the primate V1. To do this, we recorded spiking activity from V1 OD columns while macaque monkeys passively viewed grating stimuli through a custom mirror stereoscope (Figure 1A). We analyzed V1 spiking responses ($n = 502$, 25 sessions, 2 subjects) to 36 combinations

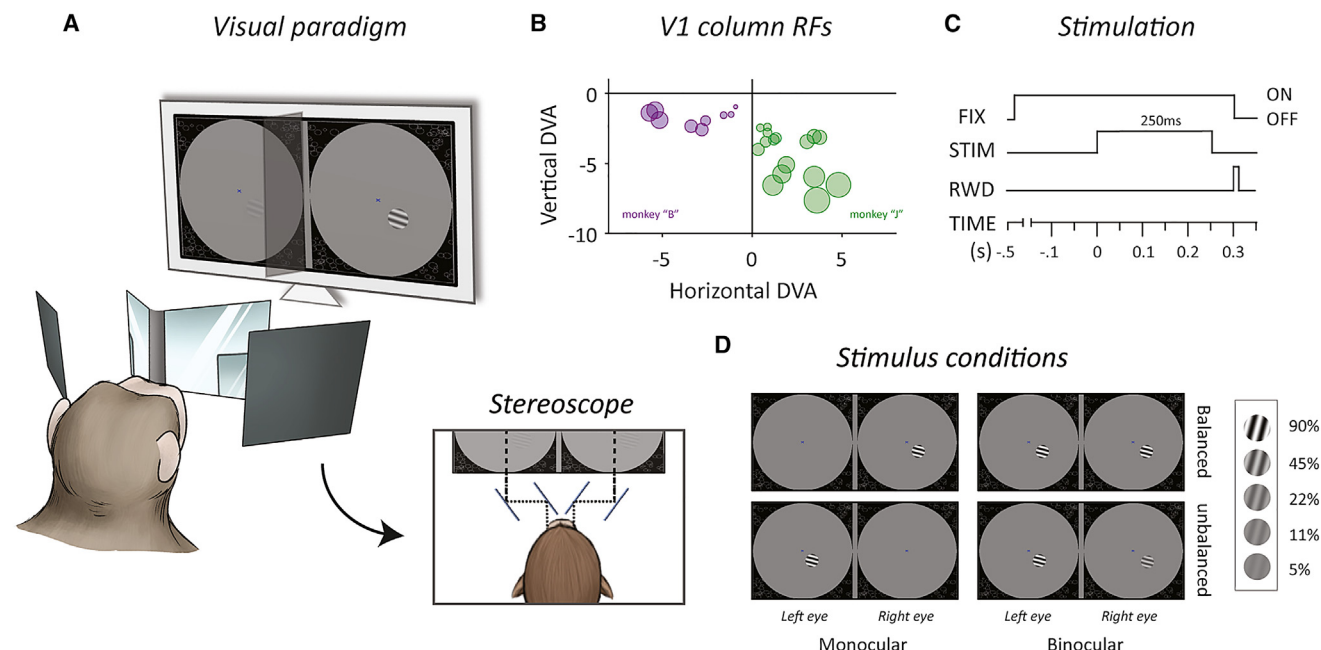


Figure 1. Experimental methodology

(A) Visual paradigm. Animals viewed all stimuli through a custom-built mirror stereoscope, such that images on the right side of the display were viewed by the right eye and images on the left by the left eye. Stimuli were presented to one or both eyes parafoveally (while fixating a central cue).

(B) V1 columnar population receptive field boundaries and extents in degrees of visual angle (DVAs) from the point of fixation (0,0). We recorded from 9 V1 penetrations from monkey B (right V1, purple) and 16 V1 penetrations from monkey J (left V1, green).

(C) Gratings appeared on screen for 250 ms, up to three times per trial with 300 ms interleaved. If fixation was held for the entire trial, monkeys received fluid reward.

(D) Stimulus conditions. Gratings were presented to one or the other eye (monocular) or both eyes simultaneously (binocular). Gratings varied in 6-octave-spaced Michelson contrast levels (0, 0.05, 0.11, 0.225, 0.45, 0.90). Binocular gratings either had the same contrast (balanced) or varied in contrast between the eyes (unbalanced).

of varying contrast across the eyes, comprising both monocular and binocular conditions (Figure 1B). Using these data, we tested for a relationship between OD, a property derived from the difference between monocular responses, and the neurons' binocular responses. We took two approaches to answering this question. First, we quantified the extent to which the neurons' dominant eye impacts binocular response formation and combination of contrast. Then, we tested whether neural models of binocular contrast combination can be improved by including information about the neurons' OD.

Determining a neuron's preference for one eye (OD)

Most neurons in the V1 respond more vigorously to stimulation of one eye over the other.²² However, even neurons located in the centers of OD columns receive inputs from neurons connected to the other eye.^{19,23–25} Here, we recorded neuronal responses from separate V1 OD columns comprising binocular neurons exhibiting a range of response preferences for either eye. To facilitate our analysis of OD's role for binocular combination, we first determined which eye was most effective at driving spiking responses for each unit ($n = 502$). Specifically, we compared responses across contrast levels between stimulation of the contra- and ipsilateral eye. Using these data, we calculated an OD index at each contrast level as the difference between the mean response u for unit i to contralateral presentations and ipsilateral presentations, divided by their sum:

$$(OD)_i = \frac{Contra(u_i) - Ipsilateral(u_i)}{Contra(u_i) + Ipsilateral(u_i)}$$

The output of this formula is a value between -1 and 1 , with -1 representing units driven more vigorously by the ipsilateral eye and 1 describing units driven entirely by the contralateral eye. We averaged the obtained values across stimulus contrast to derive a single OD index for each unit (Figure 2A). Approximately half of the neurons preferred the contralateral eye (49.2%). Because neurons varied in which eye they preferred from column to column, we organized responses to the 36 conditions by contrast shown to the "dominant" versus "non-dominant" eye (Figure 2B).

The dominant eye has a greater impact on V1 binocular response formation than the non-dominant eye

We started by investigating the relative role of each eye for binocular contrast combination. Analyzing pairs of binocular conditions with swapped contrast in the two eyes provides clues to answer this question. If OD were irrelevant for binocular contrast combination, a binocular response should be identical when one or the other eye receives higher contrast. If, on the other hand, swapping contrasts between the eyes does not lead to equivalent binocular outputs, OD needs to be taken into account to fully describe binocular contrast combination. To adjudicate between these two possibilities, we compared

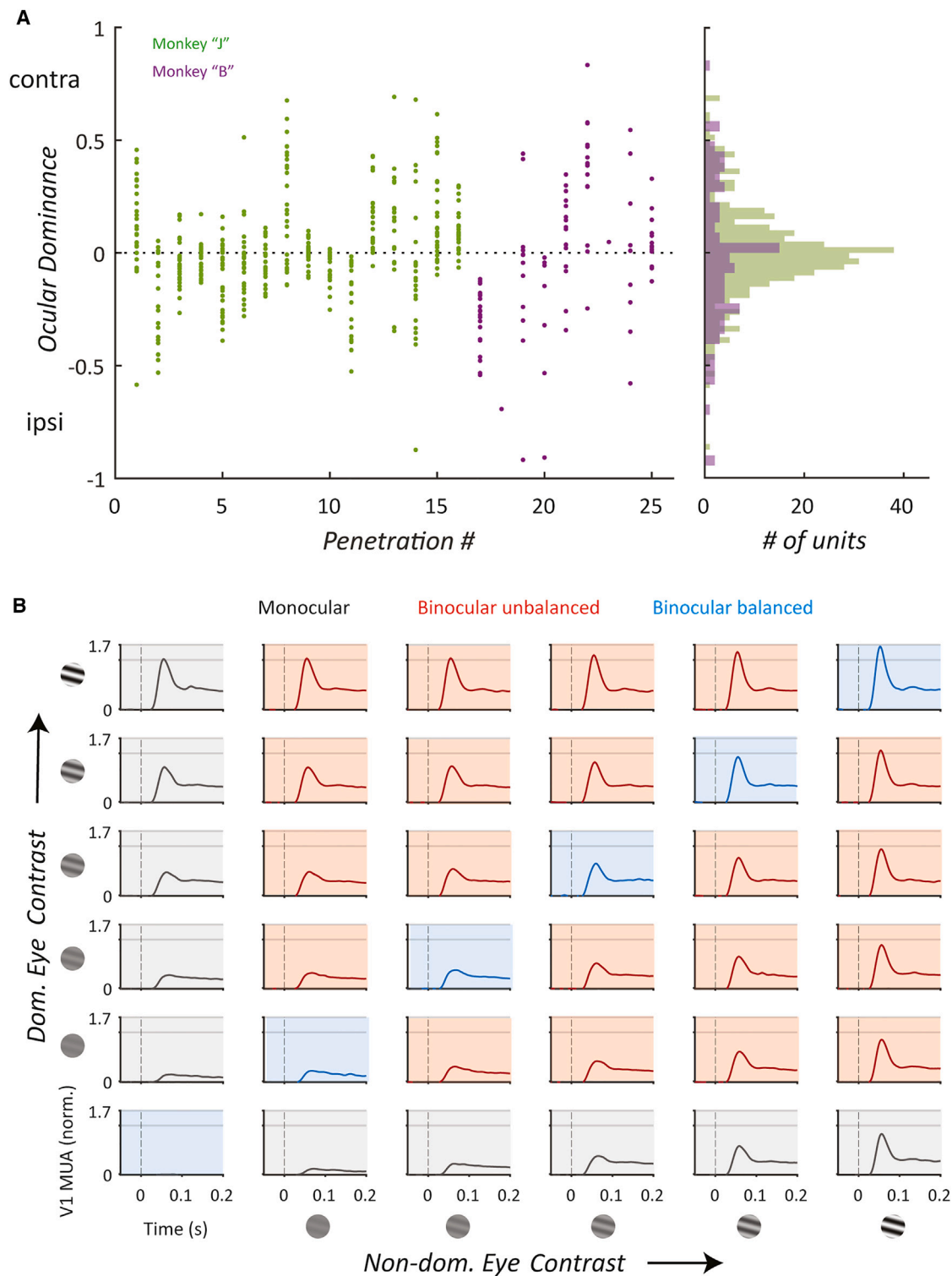


Figure 2. Determining a neuron's dominant eye

(A) Ocular dominance across penetrations ($n = 25$). Each datapoint represents a unit ($n = 502$; 404 from monkey J). Units with ocular dominance values greater than zero exhibited a response preference for the contralateral eye. Histogram shows the distribution of ocular dominance for each subject.

(B) Mean V1 responses ($n = 502$) to all 36 stimulus conditions, organized by contrast shown to the dominant and non-dominant eye. Monocular responses are plotted in gray; binocular responses to balanced gratings (same contrast) are plotted in blue; binocular responses to unbalanced gratings (different contrasts) are plotted in red.

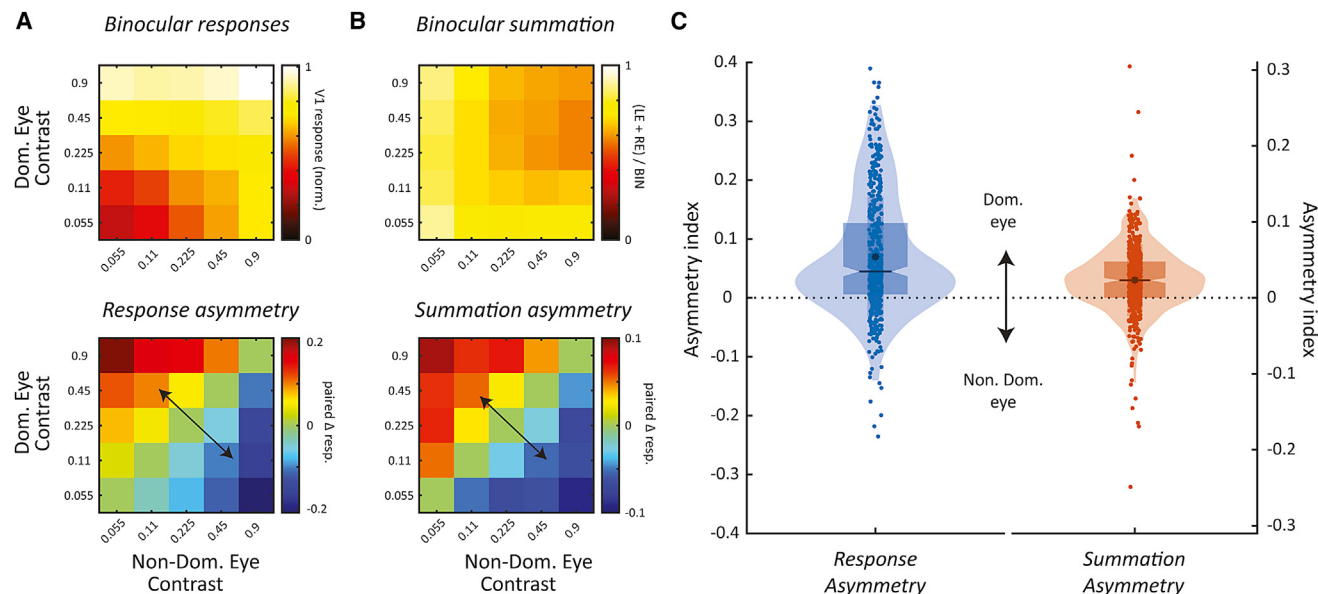


Figure 3. The dominant eye carries more weight

(A) Mean V1 binocular response matrix (average across 502 units) to balanced (diagonal) and unbalanced contrast in the two eyes (off-diagonal). The matrix is organized such that the upper left cells represent responses to binocular conditions where the contrast was always higher in the dominant eye. Below is a pairwise subtraction between upper left cells and lower right cells to assess whether contrast in the dominant eye carries more weight into the binocular responses. The resulting calculation revealed that V1 responses to unbalanced contrasts in the two eyes are greater when the higher contrast is placed in the dominant eye. (B) Mean V1 binocular summation matrix (average across units) to balanced (diagonal) and unbalanced contrast in the two eyes (off-diagonal). Summation asymmetry is plotted below, using the same procedure as described for (A). Binocular summation was greater for unbalanced contrast when the higher contrast is placed in the dominant eye.

(C) Unit-by-unit distribution of binocular response asymmetry and binocular summation asymmetry simplified to a single scalar (502 units). Values above and below zero indicate asymmetry favoring higher contrast in the dominant or non-dominant eye, respectively. Individual points represent raw data; mean indicated with an asterisk; median indicated with a horizontal black line that defines the notch in the box; bottom and top of the box indicate 1st and 3rd quartiles, respectively; bottom and top of the shaded “violin” region indicate 1st and 99th percentile, respectively.

See also [Figure S1](#).

binocular responses (trial-averaged, 50–200 ms) that received the same total contrast energy but differed in which eye (dominant versus non-dominant) received the higher contrast (e.g., 11%/45% versus 45%/11%). [Figure 3A](#) plots a summary of binocular response magnitudes (averaged across 502 units) organized by contrast in the dominant eye (upper left of the matrix) and contrast in the non-dominant eye (lower right of the matrix). Note that the magnitude of the binocular response increased as a function of total contrast energy. To assess whether contrast in the dominant eye contributed more to binocular responses than contrast in the non-dominant eye, we flipped the matrix about the diagonal and performed a subtraction between the upper and lower off-diagonal responses (i.e., paired contrast conditions). The bottom panel of [Figure 3A](#) plots the results of this calculation, which we refer to as *response asymmetry*. Finally, we computed an index of response asymmetry for each unit by taking the average delta between paired contrast conditions. Values above zero indicate a binocular response asymmetry that favors contrast placed in the dominant eye ([Figure 3C](#)). We found that paired binocular conditions with the same contrast energy but flipped contrast in the two eyes were statistically different (*Wilcoxon* $W = 107,185$, $p < 0.0001$, rank biserial correlation = 0.698). When the higher contrast was in the non-dominant eye, the binocular response tended to be less than when the same stimulus was placed in the dominant eye.

In a similar vein, we asked whether *binocular summation*, or the extent to which a binocular response can be explained as a linear combination of its monocular subcomponents, is impacted by which eye (dominant or non-dominant eye) receives the higher contrast. [Figure 3B](#) demonstrates the magnitude of binocular summation ($1 = \text{linear summation}$) across contrast conditions. We found that binocular summation was generally the greatest when the total contrast energy was low in the two eyes and systematically decreased with summed contrast energy. Like the procedure described above, we calculated summation asymmetry by flipping the matrix about its diagonal and subtracting lower and upper off-diagonal responses, then converting these delta values into an average index for each unit. Values above zero indicate a binocular summation asymmetry that favors contrast placed in the dominant eye ([Figure 3C](#)). We found that paired binocular conditions with the same contrast energy but flipped contrast in the two eyes summed contrast differently (*Wilcoxon* $W = 101,420$, $p < 0.0001$, rank biserial correlation = 0.607). When the higher contrast was in the non-dominant eye, binocular summation tended to be less than when the same stimulus was placed in the dominant eye.

OD index predicts each eye's contribution to the binocular response

So far, we have shown that contrast placed in the neurons' dominant eye carries more weight into the resulting binocular response

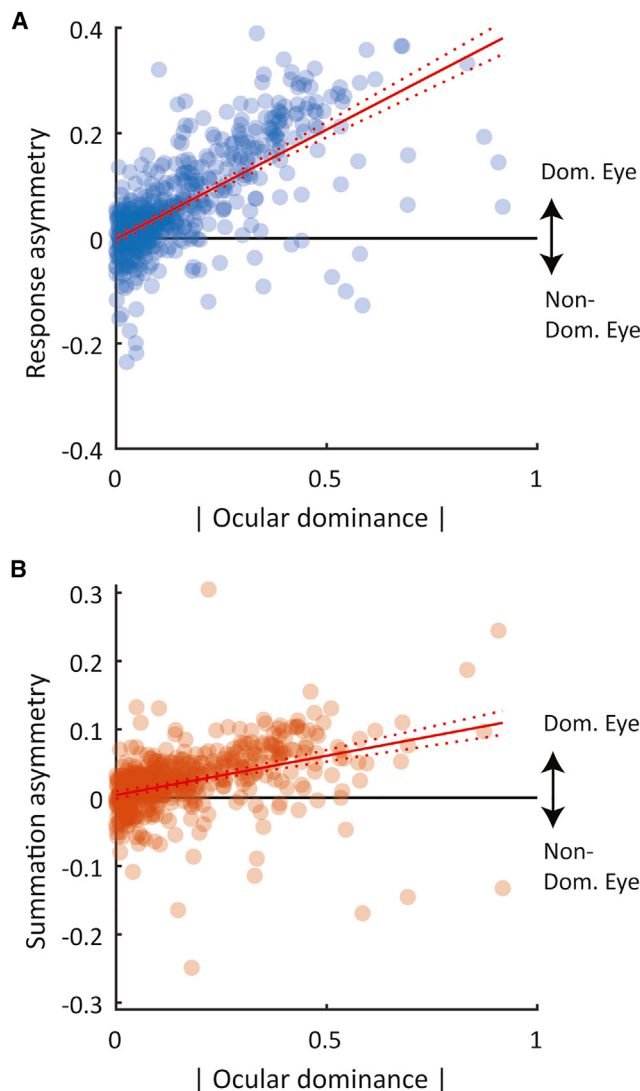


Figure 4. Ocular dominance predicts binocular response formation and contrast combination

(A) Absolute ocular dominance index plotted as a function of binocular response asymmetry index ($n = 502$ units).

(B) Absolute ocular dominance index plotted as a function of binocular summation asymmetry index ($n = 502$ units). Datapoints that fall above the zero line indicate that the neuron's binocular response or summation was greater when the higher contrast was placed in the neuron's dominant eye, respectively. Solid red line represents simple linear regression model of the asymmetry indices (y) fitted to the ocular dominance data (x); dotted line indicates 95% confidence bounds.

See also Figure S1.

than contrast placed in the non-dominant eye. Our organization of dominant and non-dominant eye was based on which eye drove monocular responses more vigorously, which is the common definition of OD. If OD impacts the formation of the binocular response, then the magnitude of OD should predict the magnitude of response and summation asymmetry (shown in Figure 2). To test this, we took the absolute value of each unit's OD to produce a value between 0 and 1, with 0 representing the most binocular units and 1 representing the most monocular units.

We then plotted the strength of absolute OD as a function of response and summation asymmetry indices. We found a relationship between both asymmetry indices and absolute OD (Figure 4). Specifically, as absolute OD increased, mean asymmetry in binocular responses (Figure 4A) and binocular summation (Figure 4B) systematically increased, respectively (adjusted $R^2 = 0.46$, $t = 20.53$, $p = 2.0798 \times 10^{-68}$; adjusted $R^2 = 0.16$, $t = 9.90$, $p = 3.2962 \times 10^{-21}$). See Figure S1 for the same analysis conducted on a small sample of well-isolated single neurons from these recording sites in V1. These findings show that a neuron's absolute OD predicts how strongly that neuron weighs contrast in the dominant eye during binocular response formation.

Modeling OD within a divisive normalization framework

Results so far suggest that OD plays a role in binocular contrast combination by modulating the effectiveness of contrast, depending on whether it is shown to the dominant or non-dominant eye. These results are in line with a recent demonstration that OD modulates stimulus *gain* in the cat striate cortex.²¹ To find out whether this type of modulation also explains our observations, we wanted to test whether computational models of binocular contrast combination can be improved by including information about a neurons' OD.

We started by modeling monocular responses to stimulus contrast within a divisive normalization framework. Divisive normalization²⁶ has been successful at explaining V1 responses to contrast and a range of other V1 phenomena.²⁷ Accordingly, normalization has been proposed to represent a canonical cortical computation.²⁸ For each unit, we measured monocular contrast response functions (CRFs) by fitting trial-averaged spiking responses to each contrast level with the Naka-Rushton equation,²⁹ allowing all parameters ($R_{\max}$, C_{50} , n) to vary freely. As expected, the parameters of the dominant and non-dominant eye CRFs differed. We hypothesized that, by parameterizing the difference between *monocular* curves, we could effectively parameterize OD as a kind of gain-control on contrast placed in the neurons' non-dominant eye. This enabled us to then test whether inclusion of OD improves the ability of *binocular* models to fit V1 binocular responses. Our first objective was to determine which parameter ($R_{\max}$, C_{50} , n), or which kind of gain-control, explains the most variance between the dominant and non-dominant eye CRFs (Figure 5A). To achieve this, we first fitted a curve to the neurons' dominant eye responses to contrast (see Figure S2 for characterization of fits to the monocular data). We then fitted a curve to the neurons' non-dominant eye responses, but this time we fixed two parameters to the dominant eye and allowed just one to vary freely. Figure 5B demonstrates the effect of allowing each parameter to vary while holding the other two constant. We found that allowing the parameter $R_{\max}$ to vary freely between the fitting dominant and non-dominant eye responses yielded the best fits (mean adjusted $R^2 = 0.92$, $SD = 0.09$) for the non-dominant eye responses (Figure 5C). This finding suggests that the predominant expression of OD within a divisive normalization framework can be described as response-gain control, or a vertical shift of the CRF along the ordinate. The same effect can be achieved by adding a weight to the contrast in the numerator of the Naka-Rushton equation (Figure 5D). Re-fitting the non-dominant again, with all parameters fixed to the dominant eye but with an additional term (w), matched the effects

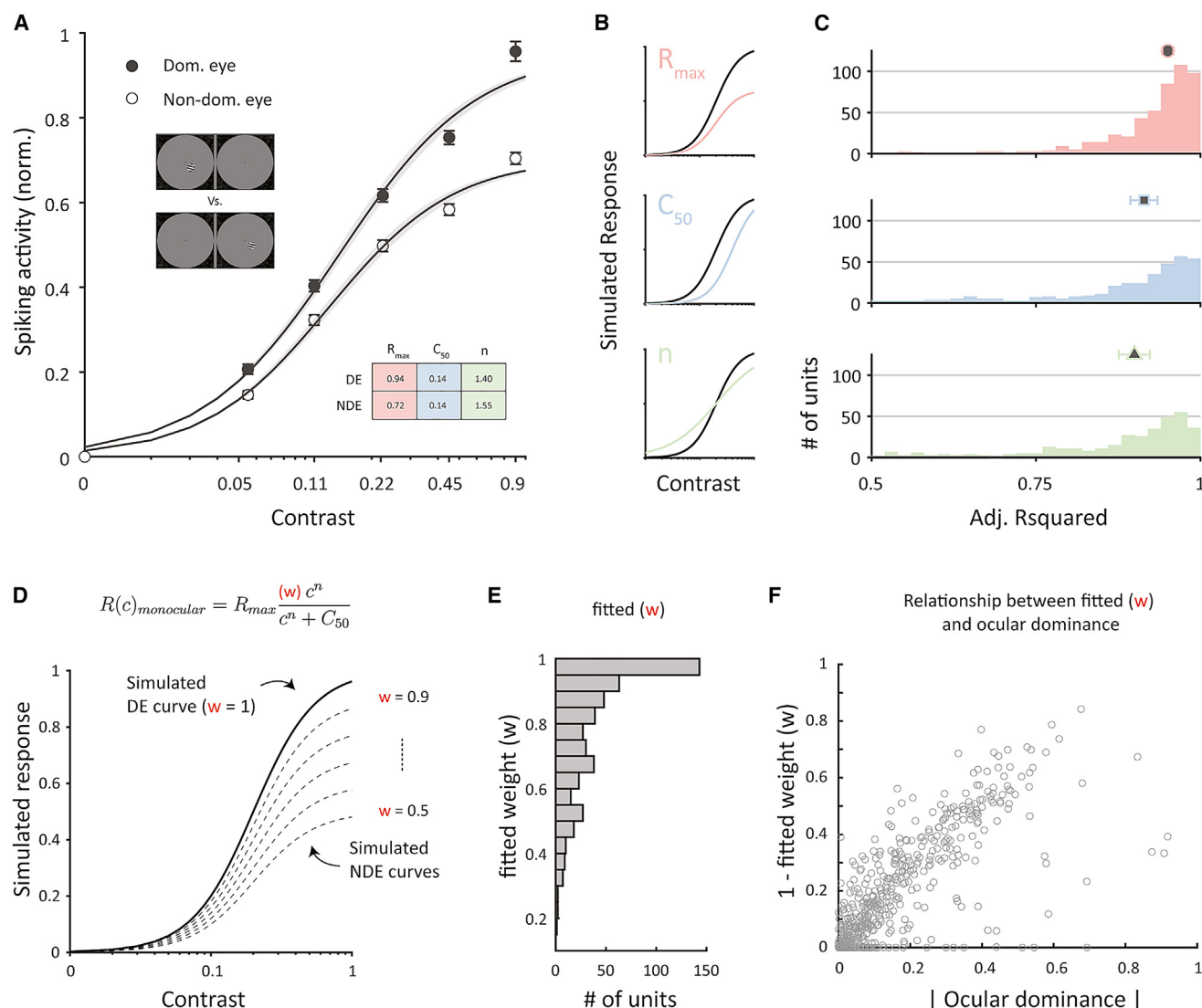


Figure 5. Modeling ocular dominance within a divisive normalization framework

(A) Mean V1 responses as a function of stimulus contrast placed in the dominant and non-dominant eye fitted with the Naka-Rushton equation. Error bars on data points refer to the SEM of the population-averaged data ($n = 502$). Mean parameters for each curve are displayed in the inset box.

(B) Effect of each parameter on the contrast response curve.

(C) Distribution of goodness of curve fit to the non-dominant eye responses, allowing just one parameter to vary at a time while fixing the other two to the dominant eye curve ($R_{\max}$, $M = 0.92$, $SD = 0.09$; C_{50} , 0.82 , $SD = 0.21$; n , $M = 0.79$, $SD = 0.24$). Colored markers indicate the mean $\pm$ SEM of each distribution.

(D) Placing a weight (w) on contrast in the numerator matches a response-gain control.

(E) Distribution of weights (w) after allowing the weight to vary while fixing all other parameters to the dominant eye curve.

(F) Relationship between neurons' fitted weight and absolute ocular dominance index.

See also Figure S2.

of $R_{\max}$ and produced a distribution of weights (Figure 5E) that correlated (adjusted $R^2 = 0.56$, $t = 25.2$, $p = 2.6770e^{-91}$) with the neuron's OD index (Figure 5F).

Accounting for OD improved computational models of binocular contrast combination

Weighting the contrast in the non-dominant eye (w) captured the effect of OD as a response-gain control within a divisive normalization framework. We next tested whether the inclusion of this weight, derived from the monocular responses of each unit, can

improve the ability of various binocular models to fit the neurons' binocular responses. We tested four simple models of binocular contrast combination that are built using the divisive normalization framework: (1) binocular average, (2) linear summation, (3) quadratic summation,³⁰ and (4) interocular normalization.³¹ Details of each model and fitting procedure can be found in the STAR Methods.

Recall that V1 binocular responses exhibited a response asymmetry, which can be visualized as a bias in response magnitude toward contrast placed in the dominant eye (Figure 6A). We

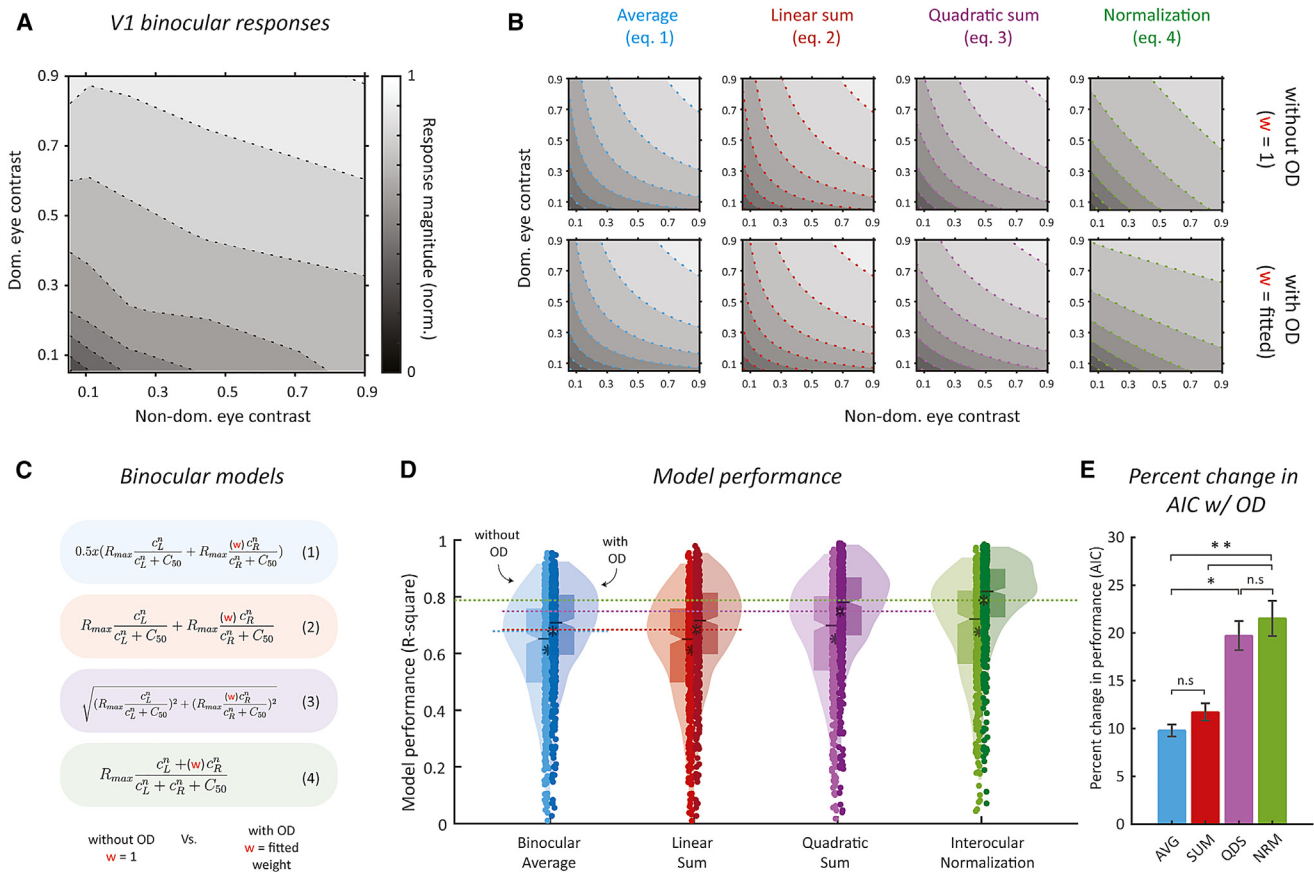


Figure 6. Weighting contrast in the non-dominant eye improves computational models of binocular combination

(A) Contour map of V1 binocular responses highlighting binocular response asymmetry.

(B) Mean model performance (fitted to mean data) demonstrates the effect that the fitted ocular dominance weight (w) has on binocular models fits.

(C) Computational models of binocular contrast combination. The weight (w) term takes on a value of 1 when testing the models without OD and takes on the individual units' fitted w (derived solely from monocular data) when testing the models with OD.

(D) Model performance with and without OD for each unit. A horizontal line is plotted through each model's mean performance with OD to aid visual comparison.

(E) The effect of adding OD information calculated as a percent change in estimated Akaike's information criteria (AIC). Generally, the inclusion of the fitted weight derived from monocular data improved the ability of binocular models to fit the binocular response data. Error bars represent SEM; ** $p < 0.01$; * $p < 0.05$.

showed that this asymmetry varies systematically with the OD of V1 neurons. By accounting for OD in each model via a weight to contrast in the non-dominant eye, we expected model performance to improve over its baseline performance (without OD). Figure 6B conveys a visual summary of model performance at the mean (averaging responses across all units) level. Notably, models that do not incorporate OD produce symmetrical response matrices and therefore cannot account for the binocular asymmetry that we observe. Figure 6D shows distributions of model performance (adjusted R^2) without (left distribution) and with (right distribution) the fitted OD weight unique to each unit. As expected, the addition of OD information generally improved the performance of each model. A Friedman test revealed there were significant differences between models equipped with OD ($\chi^2(3) = 779$, $p < 0.001$). Post hoc comparisons using the Durbin-Conover test indicated that interocular normalization ($M = 0.79$, $Mdn = 0.82$) performed slightly better than quadratic summation ($M = 0.75$, $Mdn = 0.78$; $p < .001$), and much better than averaging ($M = 0.68$, $Mdn = 0.71$; $p < 0.001$) or summing the contrast in the two eyes ($M = 0.68$, $Mdn = 0.72$; $p < 0.001$). To quantify the gain

in performance due to OD, we approximated Akaike's information criteria (AIC) for each model fit (Table 1). The formula used for this approximation can be found in the STAR Methods. We then calculated percent change in AIC within each model for each individual unit (Figure 6E). Interocular normalization received the biggest boost in performance ($M = +21.5\%$, $SD = 41.6$), followed closely by quadratic summation ($M = +19.7\%$, $SD = 34.1$). Averaging or summing contrast between the two eyes (with one eye weighted) received the smallest boosts in performance ($M = +9.79\%$, $SD = 14.3$; $M = +11.7\%$, $SD = 20.2$, respectively).

DISCUSSION

We report that binocular combination of contrast in the macaque V1 is modulated by the neurons' ocular preference. Contrast placed in the neurons' dominant eye contributed more to binocular response formation than the same contrast placed in the neurons' non-dominant eye. This asymmetry positively correlated with the OD index of neurons. We also show that accounting for this relationship between OD and binocular response

Table 1. Effect of ocular dominance weight on model performance

	Akaike's information criteria (AIC)		Mean difference	Statistic		Rank biserial correlation
	Without OD	With OD		Wilcoxon W	p	
Average (Equation 1, STAR Methods)	52.7	47.7	5.00	13,552	<0.00001	0.79
Summation (Equation 2, STAR Methods)	52.6	46.6	6.00	12,939	<0.00001	0.80
Quadratic (Equation 3, STAR Methods)	48.8	38.2	10.6	16,603	<0.00001	0.74
Normalization (Equation 4, STAR Methods)	46.7	34.0	12.7	19,114	<0.00001	0.70

asymmetry drastically improves divisive normalization models of binocular contrast combination, suggesting that OD operates as a gain-control on contrast placed in the non-dominant eye during binocular response formation.

Relation to prior literature

The existence of OD in two-eyed mammals has been a puzzle for the past six decades.³² Since their discovery, efforts have been made to understand what role, if any, OD plays in visual processing.^{17,33–36} Anatomical studies have revealed considerable variation in OD column expression across different species, with no apparent impact on visual capacity.^{5,37} Certain characteristics of OD columns, such as their size and periodicity, differ even among individuals of the same species.³⁸ Notably, some squirrel monkeys lack these columns entirely or only exhibit them in peripheral representations of V1, yet their visual abilities remain comparable to other squirrel monkeys that express them.³⁵ These observations have led to the proposition that the arrangement of OD into columns and their topographical pattern in the V1 may not serve a specific purpose for mammalian vision.³² This proposition remains uncontested. It is important to note that our data here do not speak to the existence or significance of the spatial layout of OD into columns, nor are columns a prerequisite for the effects we have described.

An alternative, but not mutually exclusive, hypothesis is that the physiological property of OD is relevant for early visual processing. Indeed, OD has been observed in mammals without any specific spatial geometry, as in rodents.³⁹ Neurophysiological studies have not delivered replicable evidence for OD's involvement in the V1's mechanisms for depth detection,^{12,16,17,40,41} at least at the level of single neurons; see Parker et al.⁴² We propose that the ocular preference of a cell influences binocular processing more generally. Studies of binocular interactions have shown that the differential strength in ocular inputs to a given cell has a bearing on the formation of binocular outputs.^{43,44} Notably, the model of Anderson and Movshon includes a neuronal population that individually combines contrast signals from each eye in a linear manner and as a group exhibit cell-to-cell variations in the slopes of their binocular interaction contours.⁴³ This cell-to-cell variation in the slopes of binocular interaction contours has been hypothesized to reflect variation in OD.⁴⁴ In this study, we find evidence for this hypothesis. Specifically, we demonstrate a relationship between the OD of V1 neurons and their binocular responses to varying interocular ratios of visual contrast, with asymmetries in both binocular response magnitude and summation that are traceable to the effectiveness of contrast placed in the dominant and non-dominant eye, separately.

The OD index contains useful information

A lingering question in the field has been to what extent the OD index, measured purely from monocular tests, can serve as a useful parameter for our understanding of early visual processing. Hubel and Wiesel first characterized OD of individual V1 cells on a scale from 1 to 7, with 1 assigned to cells that were only driven by the contralateral eye and 7 for cells that were only driven by the ipsilateral eye.^{1,45} Using this scale, the OD of cells measured in normally sighted subjects tends to show a normal distribution of OD centered on 4, suggesting that most V1 neurons respond to each eye equivalently without much selectivity for one eye or the other.¹ Subsequent studies have used different scales (–1 to 1, or 0 to 1), but a similar distribution has been consistently observed across species.^{2,12,16,17,19,20,40,46}

Concerns have been raised regarding the validity of the OD index, as classically defined. Using traditional monocular tests, a neuron's OD index is based on the relative difference in spiking response to left- and right-eye stimuli.¹ However, a neuron's discharge magnitude does not adequately capture its synaptic inputs.²⁴ Neurons must depolarize to a critical threshold to fire an action potential. So, in the case of using monocular tests for OD, synaptic inputs from one eye may not reach this threshold and thus are not represented in the firing rate of the neuron.⁴⁷ In such a case, we might mistakenly label these neurons “monocular.” Indeed, emerging evidence suggests that a majority of monocular neurons in the V1, including those in layer 4C, receive information from the other eye.^{16,19,41,47,48} Sub-threshold excitation from one of the eyes likely explains how phase-specific binocular interactions can be observed in exclusively monocular cells after monocular deprivation.⁴¹ Moreover, neurons can also receive inhibitory inputs.⁴⁹ In a simplified circuit, a monocular neuron that receives excitatory inputs from one eye and inhibitory inputs from the other (silent eye) is theoretically excited by one but suppressed by both.⁵⁰ Although suppressive phenomena in the V1 may not necessarily imply cortical inhibition,^{51,52} the relative strength of input from the two eyes is at least partially concealed with respect to what the electrode measures.

Hence, whether from sub-threshold excitation or inhibitory influence that “silences” inputs from one of the eyes, an OD index derived from spiking does not completely capture the strength or number of connections to a given cell from each eye. It then follows that the typical OD histogram, including our own here, is only a partial index of the strength of connections from each eye for a given V1 cell. This may explain, at least in part, why efforts to relate OD to other tuning characteristics of V1 cells have fallen short or produced inconsistent results.¹² However, we show that OD built from spike rates is still a useful parameter

for our understanding of binocular processing in the V1. Our results also provide an explanation for why OD is predictive of the magnitude of transient binocular facilitation¹⁹ and sustained dichoptic cross-orientation suppression.²⁰

OD as a gain control on the non-dominant eye

Gain-control is an important process that regulates neuronal activity at various stages along the primary visual pathway.^{53–56} Here, we studied a process of gain-control that activates when a stimulus is shown to both eyes versus just one eye.

The leading neural model of a binocular cell employs a linear system that sums signals from each eye before being subjected to nonlinear spike thresholding.^{26,57} This model has been informed by numerous quantitative studies in cats and monkeys that demonstrate linear-phase-specific binocular interactions in V1 neurons.^{58,59} However, strong modulation of responses as a function of interocular phase-disparity also occurs for unequal contrasts in the two eyes.^{44,59,60} For example, even when contrast is low in the eye in which the phase-shift occurs, binocular interaction is stronger than predicted by a linear system.⁵⁹ Rather, modulatory patterns are consistent with a nonlinear contrast gain-control mechanism that enhances gain when contrast is low and reduces gain when contrast is high.⁵⁵ Our finding that binocular summation of contrast is systematically reduced as a function of total contrast energy provides further support for this theory.

The divisive normalization framework explains nonlinear contrast gain-control in the V1 remarkably well⁶¹ and has been extended to explain a diverse set of neural activity across a variety of neural systems.^{31,62–66} Because of this wide applicability, divisive normalization has been proposed as a canonical cortical operation.²⁸ Recently, divisive normalization was applied to fMRI blood-oxygen-level-dependent (BOLD) responses in the V1 to explain sublinear binocular responses.³¹ We extend this finding to the level of spiking activity in the V1 and demonstrate the need to account for OD at this spatial scale to explain asymmetry in binocular summation that favors contrast in the neurons' dominant eye. Specifically, our results indicate that V1 neurons lower the gain of contrast placed in its non-dominant eye proportionate to its OD when an image is placed in both eyes simultaneously. We were able to characterize this effect as response-gain control on the excitatory drive of the non-dominant eye, rather than the normalizing suppressive pool. Our data are consistent with the notion that mixing of inputs from the two eyes to produce binocular responses in the macaque V1 are the same rules of combination that underlie many of the distinctive response features of cortical neurons.²⁷

Implications for the study of amblyopia

Amblyopia is a condition that typically appears in early childhood when there is a disruption in the normal visual input from one eye, leading to reduced vision in that eye.⁶⁷ Hubel and Wiesel replicated this condition in animals by suturing one eyelid. They found that in the striate cortex, markedly less neurons could effectively respond to visual input from the amblyopic eye.⁶⁸ Through the use of a transneuronal tracer, [3H]proline, it was discovered that the OD columns in layer 4C of the deprived eye were reduced in size, while those of the normal eye were expanded.^{3,8,69} Consequently, it was proposed that early visual deprivation

during a critical period leads to the loss of geniculate synapses onto V1 neurons dedicated to the impaired eye, thus causing amblyopia.⁷⁰ It has been noted, however, that shrinkage of OD columns does not fully account for the severe amblyopia observed following early monocular form deprivation.⁷¹ Even after eyelid suturing from 2 weeks of age, which reduces visual acuity to the level of perceiving only hand motions, the OD columns of the deprived eye still occupy approximately a third of layer 4C.⁷² One possibility is that additional anatomical changes downstream of V1 processing contribute to amblyopia.⁷¹ Alternatively, the remainder of V1 neurons receiving input from the amblyopic eye may be suboptimally integrating information between the two eyes. Our data demonstrate the relationship between V1 neurons' preference for eye and interocular combination of contrast under normal developmental conditions. Further exploration of OD in amblyopic subjects holds promise in elucidating the precise mechanisms by which imbalanced binocular processing in the V1 manifests as reduced visual acuity and compromised depth perception in those with amblyopia.

Limitations of the study

The results obtained in this study indicate that OD plays a role in the process of combining information from both eyes. This finding contributes evidence supporting one side of the ongoing debate regarding the functional significance of OD. However, it is important to note that our data do not directly address the question of why OD exists in the first place, which pertains to its potential ultimate causes. To shed light on this question, one would anticipate evidence demonstrating that binocular combination, in some aspect, functions more effectively in a system where OD is present compared with a system where it is absent.

Conclusions

It remains to be seen whether OD columns have a function related to their structure and spatial organization in the primate V1. However, we propose that the neural property of OD in the V1 is relevant for neural computations that underlie binocular integration. Continued efforts to understand the interplay between OD/binocularity, contrast gain-control, and interocular interactions are likely to lead to more useful models for how the visual system integrates monocular signals. Each of these refinements brings us closer to a complete understanding of the neural machinery that gives rise to cyclopean vision.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.cub.2023.08.019>.

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AUTHOR CONTRIBUTIONS

B.A.M., M.A.C., and A.M. designed the research. B.A.M., B.M.C., and J.A.W. performed the research. B.A.M. analyzed the data. B.A.M. prepared visualizations. B.A.M. and A.M. wrote the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

INCLUSION AND DIVERSITY

One or more of the authors of this paper self-identifies as a member of the LGBTQIA+ community. We worked to ensure sex balance in the selection of non-human subjects.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited data		
Post-processing MATLAB code and data	This paper	Zenodo: https://doi.org/10.5281/zenodo.8161772
Experimental models: Organisms/strains		
Bonnet macaque (<i>Macaca radiata</i>)	Wake Forest University, North Carolina, USA	N/A
Software and algorithms		
MATLAB, 2012-2014 and 2016A	MathWorks	https://www.mathworks.com
MonkeyLogic	David Freedman, University of Chicago; Wael Assad, Brown University	http://www.brown.edu/Research/monkeylogic/
KiloSort	Pachitariu et al. ⁸²	https://github.com/cortex-lab/KiloSort
Jamovi, v2.3.0	The jamovi project, and Sebastian Jentschke	https://www.jamovi.org/
Other		
Vector Array	NeuroNexus	http://neuronexus.com/products/neural-probes/
U-Probe, V-Probe	Plexon Inc	https://plexon.com/
Recording equipment	Blackrock Microsystems	http://blackrockmicro.com/
Eye Link II Eye Tracker	SR Research	https://www.sr-research.com/
Cold mirrors	Edmund Optics	https://www.edmundoptics.com/
Data Acquisition Board PCI-6229	National Instruments	http://www.ni.com/en-us.html
Eye-tracking Software	SensoryMotoric Instruments	https://www.smivision.com/
Photodiode	OSI Optoelectronics	http://www.osioptoelectronics.com/
Ceramic screws	Thomas Recording	https://www.thomasrecording.com/
Dental Acrylic	Lang Dental Manufacturing	http://www.orthodonticproducts.com/buyers-guide/listing/lang-dental-manufacturing-co-inc/
Microdrive	Narishige International USA, Inc.	https://usa.narishige-group.com/
Plastic Recording Chamber	Crist Instrument; custom-made	http://www.cristinstrument.com/ ; Vanderbilt University

RESOURCE AVAILABILITY

Lead Contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the Lead Contact, Alex Maier (alex.maier@vanderbilt.com).

Materials availability

This study did not generate new unique materials. All materials found in the [key resources table](#) may be obtained commercially.

Data and code availability

- All data reported in this paper has been deposited on Zenodo and is publicly available as of the date of publication. DOIs are listed in the [key resources table](#).
- All original code has been deposited on Zenodo and is publicly available as of the date of publication. DOIs are listed in the [key resources table](#).
- Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

EXPERIMENTAL MODEL AND SUBJECT PARTICIPANT DETAILS

Two adult monkeys (Monkey J, male, *Macaca mulatta*; Monkey B, male; *Macaca radiata*) were used in this study. We report on neurophysiological data recorded during 25 total penetrations (9 in Monkey B; 16 in Monkey J) of area V1. All procedures followed regulations by the Association for the Assessment and Accreditation of Laboratory Animal Care (AALAC), Vanderbilt University's Institutional Animal Care and Use Committee (IACUC), and National Institutes of Health (NIH) guidelines.

METHOD DETAILS

Surgical Procedures

Prior to data collection, animals were implanted with a custom-designed plastic head holder and a plastic recording chamber (Crist Instruments) spanning over two separate surgeries. All surgeries were performed under sterile surgical conditions using isoflurane anesthesia (1.5–2.0). Signs, including heart rate, blood pressure, SpO₂, CO₂, body temperature, and respiratory rate were monitored continuously. During surgery, the head holder and a recording chamber were attached to the skull using transcranial ceramic screws (Thomas Recording, Gießen, Germany) and self-curing dental acrylic (Lang Dental Manufacturing, Wheeling, IL). A craniotomy was performed over the parafoveal visual field representation of primary visual cortex (V1) concurrent with the position of the recording chamber. Animals received analgesics and antibiotics for postsurgical care and closely monitored by researchers, facility veterinarians, and animal care staff for at least three days following surgery.

Visual apparatus and passive fixation

Visual stimuli were presented on a VIEWPix monitor (VPixx Technologies) running at 120 Hz (22.5-inch display, resolution 1920 x 1200 pixels) with 95% display luminance and color uniformity over 95% of the display area. Animals viewed all stimuli through a custom-built mirror stereoscope, such that images on the right side of the display were viewed by the right eye and images on the left side by the left eye (Figure 1A; the monitor was divided by a black, non-reflective septum). The mirrors of the stereoscope were infrared-transparent,⁷³ enabling gaze position to be measured using infrared light-sensitive cameras (EyeLink II). To facilitate binocular fusion, matching oval apertures were displayed at the edge of each half of the screen. At the beginning of each experimental session, the stereoscope was calibrated via a behavioral task,⁷⁴ which required the animals to fixate on the same location in visual space while being cued in one eye only. Gaze position was measured for each fixation location and compared across eyes. The mirrors were considered aligned once gaze position was comparable for cuing in each eye.

Animals were trained to fixate (self-initiated) within 0.5° of visual angle of a centralized cue (blue x) appearing in both eyes. The task of the animal was to maintain this fixation while a series of sinusoidal grating stimuli were presented at a fixed location in parafoveal visual space. A variety of paradigms were employed to map receptive fields, determine receptive field tuning properties, and probe binocular combination of contrast. Each paradigm is described in detail later. Generally, trials began with fixation for 500ms before the first presentation appeared. If fixation was broken during this period, the trial was aborted. Each presentation lasted 250ms, roughly matching the average time of a typical fixation,⁷⁵ with a 300ms interval interleaved. If fixation was held for the duration of the trial (3 presentations, 1.35 seconds), the monkey received a juice reward. Alternatively, the next fixation cue appeared with a brief (1–5 s) delay. The animals were at liberty to end recording sessions at any point by halting the initiation of new trials.

Neurophysiological procedure

Experiments were conducted inside a radio frequency-shielded booth. During each recording session, a linear multielectrode array (U-/S-/V-Probe, Plexon Inc., Dallas, TX; Vector Array, NeuroNexus, Ann Arbor, MI) was inserted into V1 orthogonal to the cortical surface. The laminar probes consisted of either 24 or 32 active microelectrodes, linearly spaced 0.1 mm apart, with impedances ranging 0.2–0.8 M at 1 kHz. The probes were connected to an amplifier using an analog head stage (Blackrock Microsystems), with the shank of the probe used as the reference. Each recording session, one laminar probe was inserted into dorsal V1 through the intact dura mater using a chamber-mounted microdrive (a custom-designed modification of a Narishige International micromanipulator). Probes were adjusted in the z-plane until the majority of microelectrode contacts spanned the entire cortical thickness, from the subdural space to the white matter. Placement of the electrode relative to the cortical laminae was verified using previously

established neurophysiological criteria (e.g., CSD profile and power spectral density), as described below. Fluctuating extracellular voltages (referenced to the metallic electrode shaft) were amplified, filtered, and digitized using a 128-channel neural signal processing system (Blackrock Microsystems, Salt Lake City, UT). Two neural signals were recorded and stored for subsequent offline analysis: a low-pass filtered signal (0.3–500 Hz) sampled at 1 kHz, corresponding to the local field potential, or LFP, and a broadband (0.3 Hz–7.5 kHz) signal sampled at 30 kHz. The neural signal processing system also recorded non-neurophysiological analog signals related to the monitor refresh (i.e., a photodiode signal; OSI Optoelectronics, Montreal, Quebec) and eye position (i.e., voltage output of eye-tracking system), which were digitized and stored at 30 and 1 kHz, respectively. These time stamps and the photodiode signals were used to align the time-varying intracranial data with the onset of visual events. Trials where the animal's gaze left the central 1° radius around the fixation spot were excluded from analysis.

The relative depth of each microelectrode contact in cortex was determined from current source density and power spectral density analysis. Current source density (CSD) analysis was performed on the LFP signal using an estimate of the second spatial derivative appropriate for multiple contact points⁷⁶:

$$CSD(t, c) = - \frac{x(t, c - z) + x(t, c + z) - 2x(t, c)}{z^2},$$

where x is the extracellular voltage recorded in Volts at time t from an electrode contact at position c , and z is the electrode inter-contact distance (0.1 mm). To yield CSD in units of current per unit volume, the resulting CSD from the formula was multiplied by 0.4 S/mm as an estimate of cortical conductivity.⁷⁷

Receptive field mapping and tuning

Once satisfactory electrode placement was achieved, we used a reverse correlation-like technique to map the receptive fields of the single units under study. We first estimated the receptive field location from the audible MUA response to bar and grating stimuli that were moved across the screen while the animals fixated for juice reward. These stimuli were presented to each eye separately. We then had the animals fixate while a series of circular static random noise patches were presented monocularly and binocularly at pseudorandomized locations within a predetermined virtual grid that covered the estimated receptive field. Up to five stimuli were shown per trial, for 200 ms each with 200-ms blank periods interleaved. Stimulus size and grid spread varied depending on receptive field estimates, with each recording session typically including an initial “coarse” followed by a “fine” mapping phase of decreasing grid size. We used the resulting neurophysiological data to compute retinotopic three-dimensional receptive field matrices (RFMs)⁷⁸ to derive spatial maps of neuronal responses as a function of visual space. This procedure allowed us to identify the optimal position and size of grating stimuli for each V1 column. Figure 1B shows the putative RFs for 25 V1 penetrations (columns). Receptive fields for a given column overlapped due to the orthogonal angle of the microelectrode array to V1.

Parameters of the sinusoidal grating stimuli (i.e., orientation, phase, spatial frequency) were customized to the average tuning preferences along the V1 column each day. To do so, three 250ms monocular stimuli were interleaved with 300ms blank periods. Sinusoidal gratings pseudorandomly varied in phase, spatial frequency, and orientation^{20,79,80} and eye-of-origin (monocular and binocular presentations). All binocular presentations were presented with zero disparity (or close to zero disparity given that the monitor was flat rather than curved in the shape of the horopter) between the eyes. Upon completion of this procedure, we determined the column's mean preferred orientation, spatial frequency, and phase by statistically comparing spiking responses to each parameter ANOVA, $p < 0.05$.

Monocular and binocular stimuli

Sinusoidal gratings were presented on the monitor corresponding to the mapped population receptive field location of each electrode penetration (see Figure 1B for RF boundaries and extents). A trial consisted of three presentations, each lasting 250ms with 300ms interleaved (Figure 1C). Gratings were presented to one or the other eye using either the left or the right side of the monitor (monocular stimulation) or both eyes in conjunction (binocular stimulation). Gratings varied in Michelson contrast (e.g., 0, 0.055, 0.11, 0.225, 0.45, 0.90) across presentations (Figure 1D). Binocular presentations additionally varied in contrast between the eyes (e.g., 0.225 in the left eye, 0.45 in the right eye). In total, there were 36 stimulus conditions comprising of monocular, binocular balanced, and binocular unbalanced presentations.

Signal post-processing

Except for LFPs, all neurophysiological signals were extracted off-line from the recorded broadband signal using custom-written code in MATLAB (version 2021a). We computed an analog signal of multi-unit activity (MUA) by high-pass filtering the broadband signal at 750 Hz with a fourth-order Butterworth filter and rectifying.⁸¹ For this study, a multi-unit describes the neuronal signal extracted using the techniques described here from a single micro-electrode contact from a single penetration. In addition to the MUA signal, single-unit activity was extracted using Kilosort 2.0, an unsupervised machine-learning spike-sorting algorithm.⁸² We used the default parameters for sorting and clustering. After kilosort extracted clusters, we manually curated the clusters using Phy 2.0, following the general guidelines for ‘good’ / well-isolated unit selection based on amplitude-to-noise comparisons, inter-spike-interval violations, and stability over the recording session. Analysis of SUA is featured exclusively in Figure S1, related to Figures 3 and 4.

QUANTIFICATION AND STATISTICAL ANALYSIS

Channel selection and data formatting

We restricted our data to measurements within the cortical gray matter by eliminating channels at the top and the bottom of the electrode array that were void of MUA significantly above baseline firing. We also excluded units in V1 that did not show significant tuning to visual contrast (via a repeated measures ANOVA applied to monocular responses across 6 contrast levels, $p < 0.05$) or did not have at least 5 trials of data in all 36 stimulus conditions. Upon aligning MUA to the stimulus onset, we subtracted the baseline firing rate on a trial-by-trial basis. To enable comparisons between units, we normalized each units' visual responses to the maximum response across all the trials.

Modeling V1 responses within a divisive normalization framework

Monocular contrast response functions

Most cortical visual neurons respond nonlinearly to stimulus contrasts. This nonlinearity has been well-described by a divisive normalization function derived from the H-ratio sigmoidal function of Naka and Rushton.²⁹ We applied the nonlinear least squares approach to fit monocular responses to contrast, using the divisive normalization function:

$$R_{R_{max}, C_{50}, n, b}(c) = R_{max} \frac{c^n}{c^n + C_{50}^n} + b,$$

where R is the monocular response of the unit, R_{max} represents the projected maximum attainable response of the unit, C_{50} is the semi-saturation constant that represents the contrast at which the output is half of the maximum attainable output, n is the scaling exponent, and c is the contrast of the stimulus. The y-intercept b represents maintained activity and was fixed for individual neurons to the average activity during blank presentations (0% contrast in each eye). Each parameter was allowed to freely vary within customized limits to achieve the best fit to the data (R_{max} , [0 max response of the unit], C_{50} , [0.01 1], n , [0.5 7]).

Binocular contrast response functions

Several models have been proposed for how V1 binocular contrast combination is achieved in V1. These models typically vary in the number and position of gain-control mechanisms. We tested various binocular models built on a shared normalization framework which represent different ways for combining the two monocular contrast response functions:

One possibility is that gain-control on binocular signals is applied at a binocular stage after the two eyes are summed:

$$R(c_L, c_R) = 0.5 * \left(\frac{c_L^n}{c_L^n + C_{50}^n} + \frac{c_R^n}{c_R^n + C_{50}^n} \right) \quad \text{Equation 1}$$

Here, the binocular response is modeled as the average of the two monocular contrast response functions, akin to our visual experience when opening and closing one eye.

On the other hand, the two eyes may give rise to independent divisive normalization channels, without possibility for interocular interactions or any form of binocular gain-control. While this linear model has proven to be an unlikely candidate given inherent nonlinearities of spike generation,^{31,83–85} it is still a useful tool to assess the degree of summation of contrast in the two eyes:

$$R(c_L, c_R) = \frac{c_L^n}{c_L^n + C_{50}^n} + \frac{c_R^n}{c_R^n + C_{50}^n} \quad \text{Equation 2}$$

Yet another possibility is that the two eyes are normalized monocularly and then summed quadratically:

$$R(c_L, c_R) = \sqrt{\left(\frac{c_L^n}{c_L^n + C_{50}^n} \right)^2 + \left(\frac{c_R^n}{c_R^n + C_{50}^n} \right)^2} \quad \text{Equation 3}$$

This historical model employs gain-control at a binocular stage.³⁰ Unlike the previous model, here magnitude of gain-control scales nonlinearly with contrast.

Alternatively, the two eyes may interact before being summed, such that each eye's input exerts control over the strength of the other eyes' normalization.³¹ Unlike the other models, this model incorporates interocular interactions at a monocular stage:

$$R(c_L, c_R) = \frac{c_L^n + c_R^n}{c_L^n + c_R^n + C_{50}^n} \quad \text{Equation 4}$$

We used MATLAB (2021a)'s fit() function to fit each model to the binocular response matrix (5x5 trial-averaged (single contact) or mean ($n = 502$) responses) using the nonlinear least squares method with two independent variables (c_L , c_R). We additionally weighted each datapoint (i.e., condition) by the number of trials on a unit-by-unit basis. The fit procedure for the binocular response functions allowed each parameter (R_{max} , C_{50} , n) to freely vary to achieve the best fit to the data. We tested whether ocular dominance information improves each fit by adding a weight (w) as a 'problem' parameter/fixed variable to multiply the contrast in the numerator of the non-dominant eye's function. This weight took on a value of 1 (no effect) or that of each unit's fitted ocular dominance weight. The procedure for determining each unit's ocular dominance weight (w) is described in detail in the results section "Modeling ocular dominance within a divisive normalization framework" (Figure 5). Briefly, an ocular dominance weight (w) was fitted for each unit to minimize the sum of square residuals between datapoints on the non-dominant eye curve and the dominant eye curve.

Statistical analysis

Statistical analyses were performed via MATLAB or using the open-source statistics Jamovi software package (The Jamovi project 2022 (Version 2.3)). Our data analysis primarily comprised of paired samples tests. When assumptions of normality were violated (via Shapiro-Wilk's test), non-parametric tests were used (Wilcoxon sum rank tests and Friedman with Durbin-Conover post hoc tests). Model performance was assessed through comparing goodness of fit statistic obtained from each fit to the data (adjusted R^2). To evaluate the effect of the ocular dominance weight, we approximated Akaike's information criteria corrected for the number of observations and parameters (AICc) using the following formula:

$$AICc = n \ln\left(\frac{SSE}{n}\right) + 2k + \frac{2k(k+1)}{n-k-1} + n \ln(2\pi) + n$$

where n is the number of observations and k is the number of parameters in the model.